

Identifying, Characterizing, and Explaining U.S. Egg Price Shocks

A Step-by-Step Method for Detecting and Explaining Price Spikes

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ABSTRACT

U.S. retail egg prices are prone to sudden, sharp spikes during supply disruptions, but pinpointing exactly how much of each spike is caused by rising feed costs, avian flu outbreaks, or energy prices has proven difficult. This paper introduces a clear, step-by-step framework to detect and explain these price spikes using monthly U.S. egg price data spanning 1980 to 2026.

Our approach works by first building a model of what egg prices should look like under normal conditions, then measuring how far actual prices deviate from that expectation. We establish this baseline using an autoregressive model, one that predicts this month's price primarily from last month's price, trained on data from 2008 to 2013. From 2013 onward, we measure the gap between predicted and actual prices each month, and flag any month where that gap exceeds two standard deviations of recent residual volatility as a price shock.

Starting from nine detected shocks in the baseline model that spans from 2013 to 2026, adding controls for seasonal price patterns reduces this to five, the single biggest improvement across all model versions. This confirms that a meaningful share of apparent price volatility is simply predictable seasonal variation, not a true disruption. We then progressively introduce feed costs, flock losses from Highly Pathogenic Avian Influenza (HPAI), and natural gas prices to test how much each factor explains the remaining shocks. Feed costs and flock losses improve the model's explanatory power but do not fully account for the largest episodes. Natural gas prices do not improve model fit, likely because our training window was a period of falling gas prices following the 2008 financial crisis and does not reflect the relationship between energy costs and egg prices that emerged after 2022.

Every shock detected across all model versions is upward, consistent with supply-side disruptions being the primary driver of egg price anomalies in the U.S. The framework is general and can be applied to any monthly commodity price series where understanding the causes of price spikes is of interest.

Index Terms — egg prices, price shocks, $AR(1)$ model, avian influenza, feed costs, rolling z-score, out-of-sample residuals

I. INTRODUCTION

U.S. retail egg prices rose more than 136% between early 2022 and mid-2023, peaking at \$4.82 per dozen in January 2023 [9]. A nearly identical pattern played out during the 2014–2015 avian flu outbreak, when retail egg prices increased by over 20% in a single quarter and were more than 36% higher year-over-year at the peak of the disruption [2]. These episodes are not random.

Unlike heavily processed foods, where farm-level ingredients account for only about 4% of the final retail price [8], eggs reach the shelf with almost no processing in between. Feed costs alone make up roughly two-thirds of total egg production costs [7]. This means that when something disrupts supply at the farm level, whether a disease outbreak, a spike in grain prices, or an energy cost surge, that disruption passes through to retail prices quickly and visibly. There is very little buffer between the farm and the consumer.

Understanding why these spikes occur requires separating predictable price movements from genuine disruptions. Egg prices have three normal components: they tend to persist from month to month, they follow seasonal patterns tied to holidays and biological production cycles, and they respond gradually to input costs like feed. A price shock is what remains after these patterns are accounted for, i.e., the deviation that cannot be explained by ordinary market behavior.

This paper asks three questions: What does a price shock look like statistically? When did shocks occur in U.S. egg markets between 1980 and 2026? And what upstream factors (feed costs, avian flu outbreaks, energy prices) can explain them?

We build a simple baseline model—one that captures only the persistence of egg prices from month to month—measure how far actual prices deviate from it, and flag statistically large deviations as shocks. We then add covariates one at a time to test how much each factor explains. The result is not just an analysis of egg prices, it is a general, reproducible method for studying price shocks in any commodity market.

II. BACKGROUND

Egg price dynamics have been studied from several angles, and three findings from prior literature directly shape our modeling choices.

The first concerns how quickly farm-level cost changes reach the consumer. Leibtag [4] established that retail prices for minimally processed foods—a category that includes eggs—tend to follow input cost increases within 3 to 6 months, with Lamm and Westcott [3] showing the effect is faster when less processing separates farm from shelf. Together, these findings explain why we enter feed costs with a one-month lag rather than a longer delay. For eggs, the transmission from farm input to retail price is unusually direct and fast.

The second finding concerns the relative weights of different cost drivers. Muhammad et al. [6] found that during the 2022 price spike, feed costs accounted for 38.8% of the increase, natural gas for 38.0%, and bird losses from avian flu for 23.1%. Notably, they find that a 1% increase in bird loss is associated with a 1.5% long-run increase in retail prices, meaning flock losses punch above their weight. This cautions against treating any single factor as the dominant explanation and motivates our multi-covariate design.

The third finding is that supply restrictions do not need a biological cause to move prices. Mullally and Lusk [7] showed that California's Proposition 2, which imposed animal housing requirements on egg producers, raised retail egg prices by 17-21% on its own, demonstrating that policy-driven reductions in supply produce price effects comparable in magnitude to disease outbreaks.

Finally, Li et al. [5], using Chinese agricultural commodity data, found that upward price shocks tend to be more persistent than downward ones, with cost-factor influences roughly twice as strong at the high end of the price distribution as at the low end. We return to each of these findings in Section V to assess whether our results confirm or complicate them.

III. DATA

Our covariate selection is guided by the literature on egg price drivers. Muhammad et al. [6] identified feed costs, energy prices, and flock losses from HPAI as the three primary contributors to the 2022 price spike, together accounting for the full magnitude of the increase. We use retail egg prices as the dependent variable throughout and construct covariates for each channel.

The primary price series is monthly U.S. retail egg prices (Grade A, Large, per dozen) from the Federal Reserve of Economic Data (FRED) series APU0000708111, covering January 1980 through 2026. Small gaps are filled by linear interpolation. All series are log-transformed prior to modeling so that residuals can be interpreted as approximate percentage deviations and variance is more stable across time.

On the cost side, feed is the largest single input, comprising roughly two-thirds of total egg production costs [7]. We use the Bureau of Labor Statistics (BLS) Producer Price Index (PPI) for prepared animal feeds (series WPU02930102), available from 1989. This index reflects what producers actually pay for compound feed rather than raw commodity prices, making it a more direct measure of cost pressure than a raw grain index. Energy costs introduce a separate channel: geopolitical disruptions, such as the 2022 Russia-Ukraine war, transmit to egg prices via fuel costs for heating, ventilation, and transport rather than through feed grain prices. We capture this using the Henry Hub natural gas spot price from the FRED database (series MHHNGSP), available from 1997.

On the supply side, Highly Pathogenic Avian Influenza (HPAI) outbreaks reduce the number of laying hens, directly limiting supply. The primary variable is monthly layer losses from United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) surveys, log-transformed as $\log(\text{layer_losses} + 1)$ to stabilize variance and prevent $\log(0)$ errors when losses are zero. A single missing observation in October 2025 was filled by linear interpolation. Layer losses data from USDA NASS is available from 2013 onward, which establishes the start of the evaluation window. The choice of 2013 as the common evaluation start also coincides with the period of heightened HPAI activity and market volatility that motivates this study, ensuring the evaluation window captures the major outbreak episodes of

2015–2016, 2022, and 2024–2025. Earlier data exists for egg prices and feed costs, but the full set of covariates is only jointly available from this point forward.

IV. METHODS

The goal of the modeling framework is not to maximize predictive accuracy but to systematically separate normal egg price behavior from unexpected deviations. We achieve this by building a sequence of models of increasing complexity, where each version adds one explanatory factor, and interpreting the residuals of each as a measure of what remains unexplained.

The baseline model (v0) captures month-to-month price persistence, which is the minimum structure needed to define normal behavior. Exploratory analysis confirms that egg prices are non-stationary in levels (Augmented Dickey-Fuller test, $p \approx 0.95$) but stationary in first differences ($p < 0.001$). In practical terms, egg prices do not hover around a fixed long-run average—they drift upward over time—but month-to-month changes are stable and predictable. This drift also explains why baseline residuals tend to be positive on average: a model trained on 2008–2013 prices will systematically underpredict the higher levels that follow, which is why we standardize residuals using a rolling z-score rather than comparing them in raw form. We therefore model log prices as an AR(1) process:

$$\log(\text{price}_t) = c + \phi \cdot \log(\text{price}_{t-1}) + \varepsilon_t$$

where ϕ captures month-to-month price persistence and ε_t is the residual.

All models are estimated on a fixed 2008–2013 training window using [AutoReg](#) method from [statsmodel](#), with walk-forward out-of-sample predictions from January 2013 onward. The training window is chosen to reflect the modern egg market regime while avoiding the low-price, low-volatility period prior to 2008 that would bias coefficient estimates. Importantly, the AR(1) coefficient is estimated once on this training window and held fixed throughout. It is never re-estimated on the data it predicts. This ensures that the model's predictions are not influenced by the very shocks we are trying to detect

Starting from this baseline, we build four extensions by progressively adding covariates: monthly seasonal dummies in v1 (January omitted as the reference month), log feed PPI lagged one month in v2, log layer losses lagged one month in v3, and log natural gas price lagged one month in v4. All parameters are estimated jointly via ordinary least squares (OLS). Each covariate enters as a log-transformed, one-month lagged value. Log-transforming places all variables on a consistent scale and makes coefficients interpretable as elasticities, i.e., a one percent change in an input translates directly to an estimated percent change in egg prices. The one-month lag reflects the short passthrough window documented by Leibtag [4] and Lamm and Wescott [3], and ensures the model uses only information available at time $t-1$, as it would in real-world application. For layer losses specifically, the one-month lag also reflects the underlying biological mechanism: when hens are culled following an outbreak, the reduction in supply takes several weeks to translate into higher retail prices.

Since the purpose of the framework is shock detection rather than in-sample fit, all residuals are computed out-of-sample via a walk-forward loop over the 2013–2026 evaluation window. Had the model been tested on the same data it was trained on, it would appear more accurate than it truly is and genuine shocks could go undetected.

Residuals are then standardized using a rolling 12-month z-score:

$$z_t = (\varepsilon_t - \mu_t) / \sigma_t$$

where μ_t and σ_t are the rolling mean and standard deviation computed over the trailing 12 months, with a minimum of 12 observations required.

The 12-month window captures one full seasonal cycle of residual volatility, providing a stable local variance estimate without being long enough to absorb structural shifts in market conditions. The rolling mean subtraction corrects for local drift; without it, the long-run upward trend in egg prices would inflate z-scores in later periods and bias detection toward recent months. A known tradeoff of this normalization is discussed in Section VI. A month is flagged as a shock when $|z| \geq 2$, distinguishing genuinely anomalous months from routine fluctuations.

V. RESULTS

A. Model Comparison

Table I summarizes residual statistics and shock counts for all five models, evaluated over the common 2013–2026 window.

TABLE I — Model Comparison Summary

Model	Covariates	Res. STD	Res. Mean	Shocks
v0	AR(1) only	0.116	0.041	9
v1	+ Seasonality	0.102	0.031	5
v2	+ Log feed PPI, lag 1	0.115	0.039	5
v3	+ Log layer losses, lag 1	0.119	0.046	6
v4	+ Log natural gas price, lag 1	0.195	0.116	9

Figure 1 shows the baseline (v0) fit against actual prices. Figure 2 shows out-of-sample residuals for all five model versions stacked for comparison.

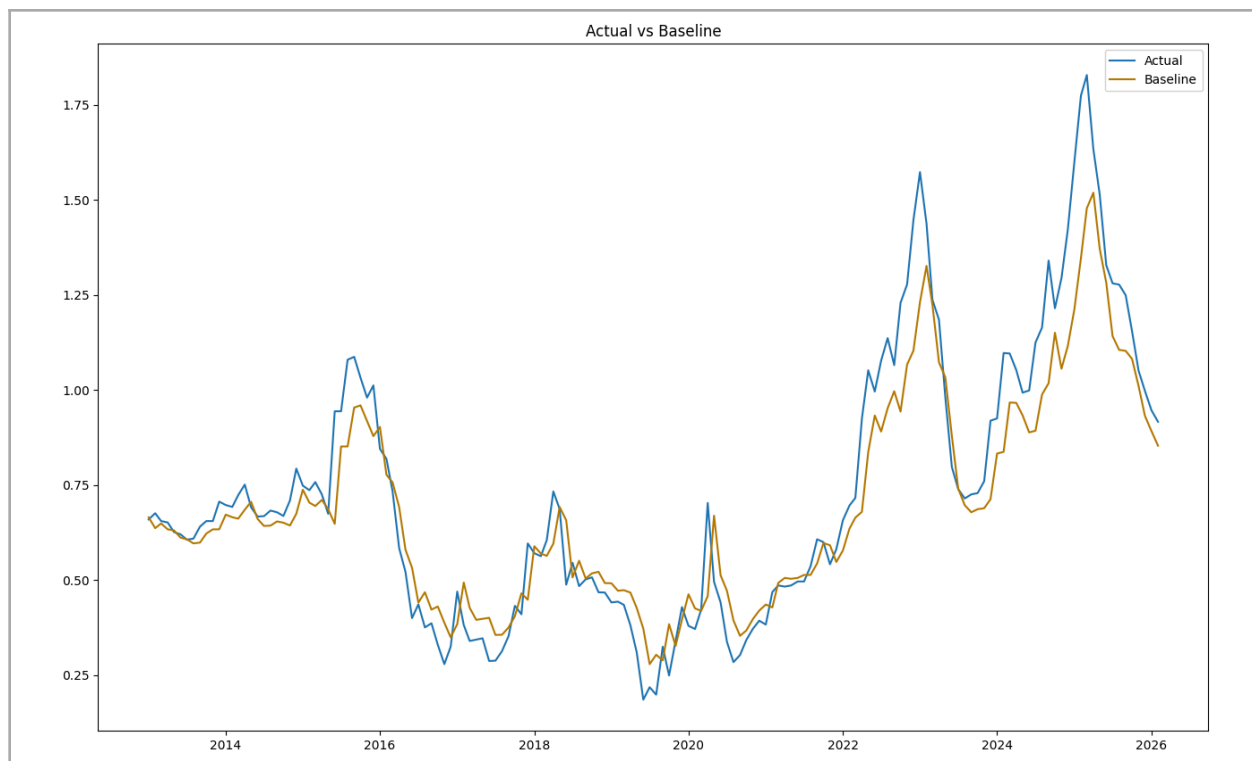


Fig. 1. Actual egg prices vs. AR(1) baseline (v0). Large deviations between the two lines correspond to shock episodes.

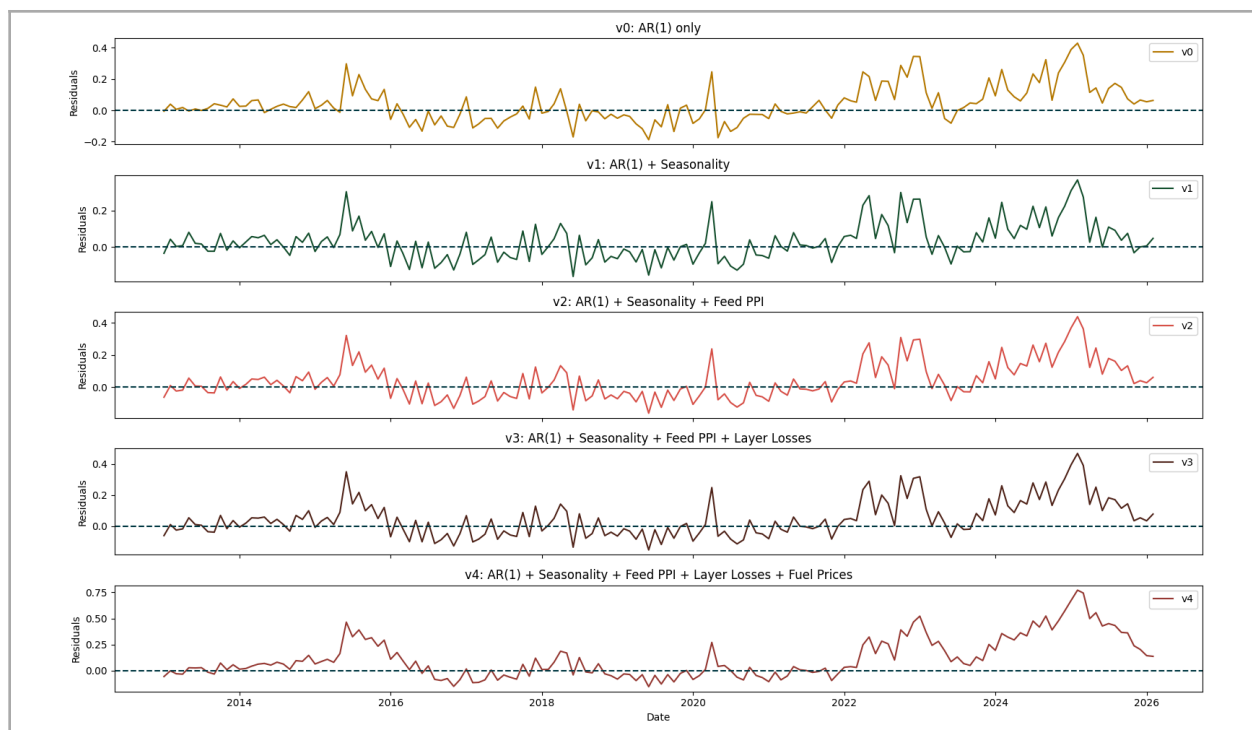


Fig. 2. Out-of-sample residuals for all five model versions over the 2013–2026 evaluation window. Each panel shows how residual structure changes as covariates are added.

The most meaningful improvement across all model versions comes from the v0 to v1 transition. Adding seasonal dummies reduces the mean residual from 0.041 to 0.031 and eliminates four shock detections that were seasonal artifacts rather than genuine price anomalies. Put simply, a meaningful share of what looked like unusual price behavior in the baseline was just predictable seasonal variation, such as holiday demand, biological production cycles, that the AR(1) alone could not account for.

The v1 to v2 and v2 to v3 transitions do not show consistent improvement in either residual standard deviation (STD) or mean. This is expected: The post-2013 evaluation window contains periods of genuine supply-side volatility that no single cost-based covariate fully explains. Residual standard deviation is therefore best interpreted as a measure of remaining unexplained volatility rather than model quality in isolation. The mean residual is the more informative metric: It captures systematic bias and tells us whether adding a covariate reduces the model's tendency to consistently under-predict prices.

The v4 results represent a clear deterioration relative to all prior versions, with residual standard deviation rising to 0.195, mean residual to 0.116, and shock count returning to 9. The natural gas covariate does not improve fit. The 2008–2013 training window coincides with a period of falling gas prices following the financial crisis, a period that does not reflect the transmission mechanism active after 2022, where energy costs and egg prices rose together. As a result, the gas coefficient is estimated on a structurally unrepresentative sample. Notably, the correlation between feed PPI and log gas price is -0.10 across the full series, ruling out multicollinearity as an explanation for the deterioration.

B. Feed Cost Passthrough

The v2 feed coefficient is +0.436, meaning a 10% unexpected increase in feed costs is associated with approximately a 4.4% increase in egg prices one month later. This is directionally consistent with both Lamm and Westcott [3] and Leibtag [4], who document rapid cost passthrough for minimally processed foods. Table II summarizes the key v2 parameter estimates.

TABLE II — Selected v2 Parameter Estimates

Parameter	Estimate	Interpretation
ϕ , AR(1) persistence	0.938	Strong month-to-month inertia under normal conditions
Feed PPI coefficient, lag 1	+0.436	10% unexpected feed cost increase $\rightarrow$ ~4.4% egg price increase one month later

Two features of these estimates are worth highlighting. First, the AR(1) persistence coefficient of 0.938 is close but below 1, which confirms that eggs prices are strongly inertial (this month's price is heavily anchored to last month's), but do eventually mean-revert rather than drift without bound. In practical terms, this means that once a shock pushes prices above the baseline, they

tend to remain elevated for several months before gradually returning to expected levels. Second, the feed PPI coefficient of +0.436 indicates that the feed cost shocks are only partially passed through to retail prices within one month. A full passthrough would imply a coefficient near 1; the estimate of 0.436 suggests that roughly half of an unexpected feed cost increase reaches the consumer within the first month, with the remainder likely absorbed gradually over subsequent periods or partially offset by other market adjustments. Together, these two parameters describe an egg market that is slow to respond to disruptions and slow to recover from them. This dynamic helps explain why shock episodes, once triggered, tend to persist across multiple months rather than resolving quickly.

C. Shock Episodes

All detected shocks across all model versions are upward. The absence of downward shocks is itself a meaningful finding: egg price anomalies in this sample are driven by supply disruptions that push prices above the model's expectation, not by demand collapses that push them below. When supply contracts, prices spike; when supply recovers, prices normalize rather than overcorrect. This is consistent with the supply-side framing in Muhammad et al. [6] and with the farm-share argument established in the introduction. Eggs have almost no buffer between farm and shelf, so supply shocks pass through quickly and fully.

The v1 shock count of 5 represents the clearest set of genuinely anomalous episodes after removing seasonal artifacts. Figure 3 shows these detected shocks plotted against the actual price series. The episodes correspond to three distinct periods visible in the residual plots as sharp positive spikes: the 2015-2016 HPAI outbreak, the COVID-era supply disruption in 2019-2020, and the sustained 2022-2025 period driven by a combination of HPAI, feed cost inflation, and energy prices.

Figure 4 shows the equivalent shock detection using v3, after adding layer losses. The persistence of shocks through v2 and v3 (counts remaining at 5 and 6 respectively) indicates that feed costs and layer losses explain part of the residual variance but do not fully account for the magnitude of the largest episodes.

The persistence is also consistent with the AR(1) structure of the model. After the 2015 HPAI shock pushed prices sharply higher, the lagged price term carried that elevated level forward into subsequent predictions. Prices corrected faster than the AR(1) anticipated, producing a brief downward overshoot in 2016-2017. This illustrates a core limitation of autoregressive models when applied to temporary supply disruptions: they struggle to distinguish a one-time shock from a lasting shift in the price level.

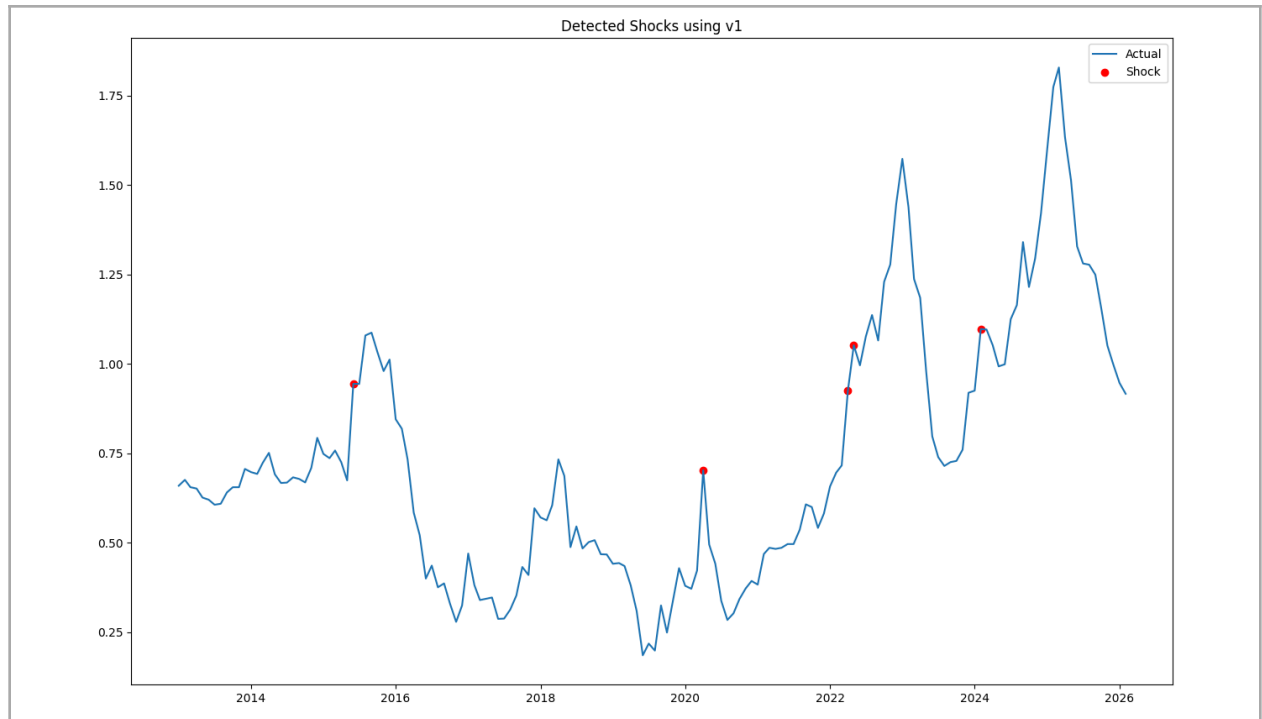


Fig. 3. Egg price shock episodes detected using v1 (AR(1) + seasonality). Red dots indicate months where $|z| \geq 2$.

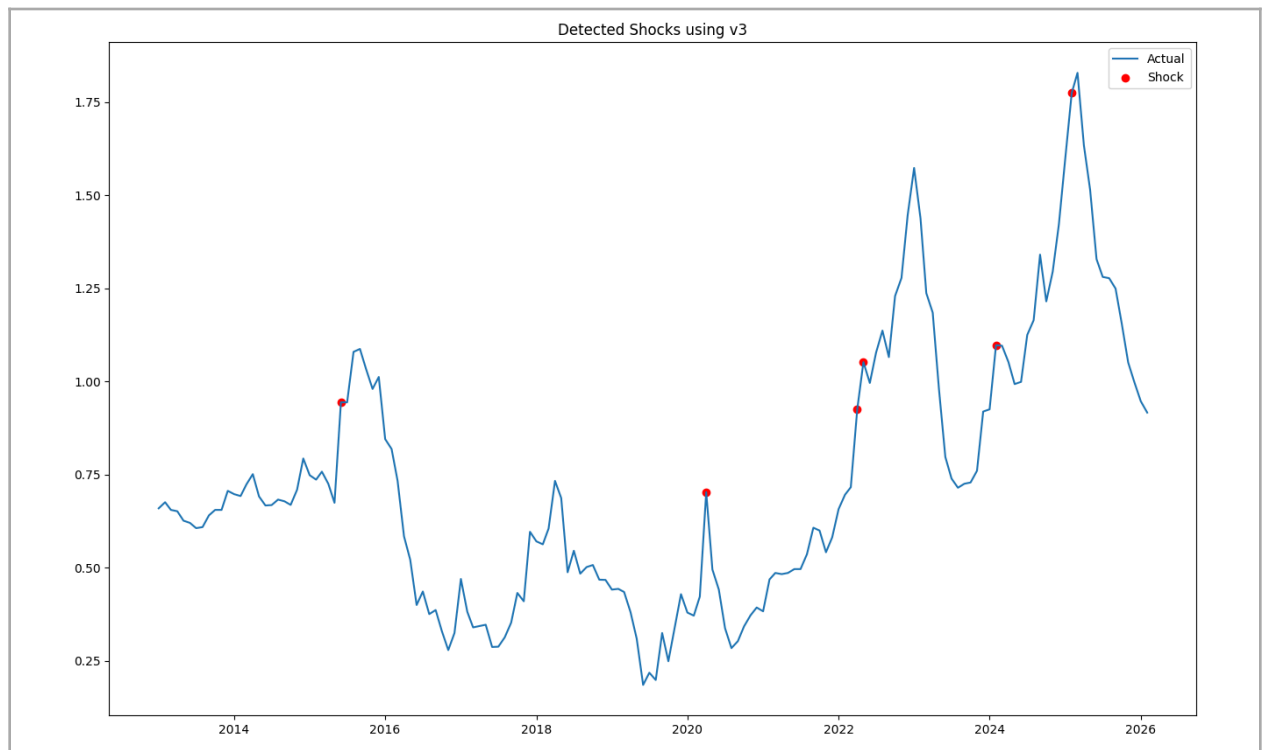


Fig. 4. Egg price shock episodes detected using v3 (AR(1) + seasonality + feed PPI + layer losses). Shocks that persist relative to v1 represent episodes not explained by feed costs or layer mortality.

VI. DISCUSSION

Our results largely confirm the supply-side picture established in the literature, with some important nuances worth unpacking.

The +0.436 feed coefficient confirms that unexpected feed cost increases propagate to retail prices within one month, consistent with the short passthrough windows documented by Lamm and Westcott [3] and Leibtag [4]. For a commodity as minimally processed as eggs, this speed of transmission is unsurprising; there is simply not enough distance between the farm and the shelf to absorb cost increases before they reach the consumer.

The v4 finding that natural gas does not improve model fit warrants closer attention. The theoretical motivation is sound: geopolitical disruptions raise fuel prices, which raise farm energy and transport costs, which pass through to egg prices. This channel is conceptually distinct from feed PPI, which captures grain input costs. The failure is not conceptual but empirical. The 2008-2013 training window coincides with a period of falling gas prices following the financial crisis; a period that does not capture the transmission mechanism that became active at scale after 2022, when the Russia-Ukraine war drove energy and egg prices upward together. A model trained on data that includes the post-2022 period would likely produce a more reliable gas coefficient. This points to a broader limitation of fixed training windows: When the structural relationship between two variables shifts over time, a coefficient estimated on pre-shift data may not only fail to help, it can actively distort predictions.

The most important methodological consideration concerns the rolling z-score normalization. By subtracting the rolling mean from residuals, the framework adjusts for local drift and prevents the long-run upward trend in egg prices from inflating shock counts in later periods. The tradeoff, however, is that slow-building sustained price increases can be partially absorbed into the rolling mean window before they are flagged. The 2022-2023 sustained price surge, the largest in the sample, shows fewer flagged shock months than the sharper but shorter 2015 episode, precisely because the gradual rise was incrementally absorbed into the rolling baseline. Researchers applying this framework should consider whether the shocks of interest are sharp and brief, where rolling normalization helps by filtering out background drift, or sustained and gradual, where it may dampen detection sensitivity.

Finally, the absence of downward shocks in our result appears to run counter to Li et al. [5] who find that upward shocks are more persistent than downward ones, which implies downward shocks should also be present. The difference likely reflects structural features of U.S. egg markets. During supply shortfalls, imports can partially offset domestic production losses, compressing the magnitude and duration of price spikes. This same mechanism may also prevent prolonged downward deviations: When supply recovers, prices normalize relatively quickly rather than overcorrecting below baseline. The asymmetry Li et al. [5] observe in Chinese agricultural markets may therefore not generalize to a market with greater trade flexibility and a more developed supply chain infrastructure.

VII. LIMITATIONS AND FUTURE WORK

Like any modeling framework, this one carries assumptions that bound its conclusions, and being transparent about them is essential to interpreting the results correctly.

The first concerns the fixed AR(1) coefficient. Because the coefficient is estimated on the 2008–2013 training window and held constant throughout, it implicitly assumes that the underlying dynamics of egg price persistence did not change after that period. The post-2020 volatility, driven by a convergence of pandemic supply disruptions, unprecedented HPAI outbreaks, and energy price shocks, suggests that assumption may not hold. If egg price dynamics shifted structurally after 2020, the fixed coefficient could introduce systematic bias in later predictions, making some shock episodes harder to detect and others easier.

The second concerns sample length. The 2013–2026 evaluation window, while sufficient to capture the episodes analyzed here, is relatively short for drawing high-confidence conclusions about rare, large-magnitude events. With only a handful of major shock episodes in the sample, it is difficult to distinguish patterns that reflect stable structural features of the egg market from those that are specific to this particular window of observation.

The third concerns attribution. As Muhammad et. al. [6] note, feed costs, HPAI outbreaks, and energy prices did not arrive separately during the 2022 spike, they co-occurred and likely amplified one another. The associations reported here are correlational, not causal, and the individual contribution of each factor cannot be cleanly separated within this framework. Readers should interpret the covariate coefficients as partial associations rather than causal estimates.

These limitations point toward natural extensions. Allowing the AR(1) coefficient to update over time through a rolling estimation window or a state-space model would better capture structural shifts in price dynamics as they occur. Applying the framework to real-time data as new observations arrive would move it toward a genuinely dynamic early-warning system for commodity price disruptions. Incorporating additional covariates such as flock inventory levels, import volumes, or policy dummies for animal welfare legislation could further decompose the unexplained residual variance and bring the framework closer to a complete account of egg price behavior.

VIII. CONCLUSION

This paper develops and applies a transparent, step-by-step framework for identifying and attributing price shocks in commodity markets. Applied to U.S. egg prices from 1980 to 2026, the framework detects a set of upward shock episodes between 2013 and 2026, all consistent with known supply disruptions — avian flu outbreaks, pandemic-era supply chain stress, and the sustained cost pressures of 2022–2025.

The results carry several clear takeaways. Seasonality is the single most important factor to control for before declaring a price movement anomalous; failing to account for predictable seasonal patterns inflates apparent shock counts and can mislead attribution efforts. Feed costs pass through to retail prices quickly and measurably, consistent with the short transmission windows documented in the literature for minimally processed foods. Layer losses from HPAI outbreaks add explanatory structure but do not fully account for the largest shock episodes, suggesting that no single factor operates in isolation during major disruptions. And natural gas prices, despite a sound theoretical motivation, fail to improve model fit in this training window; a

finding that itself has interpretive value, pointing to the limits of fixed training windows when structural transmission mechanisms shift over time.

The framework is generalizable. Any monthly commodity price series can be analyzed using the same core steps: fit an AR(1) with seasonal controls as a baseline, compute walk-forward out-of-sample residuals, and standardize against recent volatility to flag anomalous months. The method is interpretable, reproducible, and straightforwardly extensible with additional covariates as data availability permits. As egg markets, and commodity markets more broadly, continue to face compounding disruptions from disease, climate, and geopolitical shocks, frameworks that can cleanly separate signals from noise will only become more useful.

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APPENDIX

A. Model Equations

The five model versions in explicit form:

v0 (AR(1) baseline):

$$\log(\text{price}_t) = c + \varphi \cdot \log(\text{price}_{t-1}) + \varepsilon_t$$

v1 (AR(1) with seasonal controls):

$$\log(\text{price}_t) = c + \varphi \cdot \log(\text{price}_{t-1}) + \sum \beta_m \cdot D_m(t) + \varepsilon_t$$

v2 (v1 with feed costs):

$$\log(\text{price}_t) = c + \varphi \cdot \log(\text{price}_{t-1}) + \sum \beta_m \cdot D_m(t) + \gamma \cdot \log(\text{FeedPPI}_{t-1}) + \varepsilon_t$$

v3 (v2 with layer losses):

$$\begin{aligned} \log(\text{price}_t) = & c + \varphi \cdot \log(\text{price}_{t-1}) + \sum \beta_m \cdot D_m(t) + \gamma \cdot \log(\text{FeedPPI}_{t-1}) \\ & + \delta \cdot \log(\text{LayerLosses}_{t-1} + 1) + \varepsilon_t \end{aligned}$$

v4 (v3 with natural gas prices):

$$\begin{aligned} \log(\text{price}_t) = & c + \varphi \cdot \log(\text{price}_{t-1}) + \sum \beta_m \cdot D_m(t) + \gamma \cdot \log(\text{FeedPPI}_{t-1}) + \\ & \delta \cdot \log(\text{LayerLosses}_{t-1} + 1) + \eta \cdot \log(\text{GasPrice}_{t-1}) + \varepsilon_t \end{aligned}$$

Where:

- c is the intercept,
- φ is the AR(1) persistence coefficient,
- $D_m(t)$ are monthly seasonal dummies with January omitted as the reference month, and
- ε_t is the residual at time t .

B. Estimation Procedure

All model parameters (the AR(1) persistence coefficient, seasonal intercepts, and covariate coefficients) are estimated jointly via ordinary least squares (OLS) on the fixed 2008–2013 training window using statsmodels' AutoReg. HC3 heteroskedasticity-robust standard errors are used for covariate regressions, because residual variance is visibly larger during shock periods than during calm periods, violating the constant-variance assumption of standard OLS inference.

C. Z-Score Definition

Residuals are standardized using a rolling 12-month z-score:

$$z_t = (\varepsilon_t - \mu_t)/\sigma_t$$

where μ_t and σ_t are the rolling mean and standard deviation computed over the trailing 12 months, with a minimum of 12 observations required. The rolling mean subtraction normalizes for local drift in residuals, preventing the long-run upward trend in egg prices from inflating z-scores in later periods. A known tradeoff is that sustained multi-month price increases may be partially absorbed into the rolling mean window, potentially dampening detection of slow-building shocks. A month is flagged as a shock when $|z| \geq 2$.

D. Data Sources

- Egg prices: FRED APU0000708111, monthly, 1980–2026
- Feed PPI: BLS WPU02930102, monthly, from 1989
- Natural gas price: FRED MHHNGSP (Henry Hub spot), monthly, from 1997
- Layer losses: USDA NASS monthly surveys (usable data 2008 onwards)

E. Implementation

All analysis was conducted in Python using NumPy, Pandas, Matplotlib, and Statsmodels. The key pipeline steps are as follows:

1. Load and clean all raw series; apply log transformation to egg prices, feed PPI, layer losses, and natural gas prices.
2. Fit the AR(1) baseline with seasonal dummies on the 2008–2013 training window via statsmodels' AutoReg.
3. Compute walk-forward out-of-sample residuals from January 2013 onward.
4. Compute rolling 12-month z-scores; flag shock months where $|z| \geq 2$.
5. Add covariates sequentially as log-transformed, one-month lagged values to produce model versions v2 through v4.
6. For each model version, generate the following diagnostic plots: actual vs. predicted prices, residuals and z-scores over time, and detected shock episodes against the actual price series.
7. Summarize residual standard deviation, residual mean, and shock count for each model version in a comparison table.